Case Study 3: Data Audit Report

Wikipedia Web-Scraped Architecture Dataset

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Property	Value
Total Records	150
Total Columns	7
Numeric Columns	1 (year)
Categorical Columns	6
Architects Covered	7 unique architects
Building Types	55 unique categories

1. Dataset Description

This dataset was collected by web scraping Wikipedia pages of notable buildings designed by famous architects. It contains 150 records with 7 columns: **building_name**, **architect**, **location**, **year** (construction year), **building_type**, **coordinates**, and **wiki_url**. The dataset covers 7 architects including Zaha Hadid, Richard Rogers, Nicholas Grimshaw, Alfred Waterhouse, Hopkins Architects, Richard Seifert, and Berthold Lubetkin. Construction years range from 1918 to 2027.

Column	Data Type	Non-Null Count	Description
building_name	object (str)	150 / 150	Name of the building
architect	object (str)	150 / 150	Architect name
location	object (str)	107 / 150	Building location
year	float64	76 / 150	Construction year
building_type	object (str)	150 / 150	Type/category of building
coordinates	object (str)	136 / 150	Lat/Long coordinates
wiki_url	object (str)	150 / 150	Wikipedia URL

2. Data Consistency Check

2.1 Data Type Consistency

Each column was checked for mixed data types. The results show that all columns have consistent data types. String columns (building_name, architect, location, building_type, coordinates, wiki_url) contain only **str** and **NoneType** (for missing values). The year column contains only **float** values. No type inconsistency was detected.

Column	Detected Types	Consistent?
building_name	str	Yes
architect	str	Yes
location	str, NoneType	Yes (NoneType = missing)
year	float	Yes
building_type	str	Yes
coordinates	str, NoneType	Yes (NoneType = missing)
wiki_url	str	Yes

2.2 Categorical Variable Check

The **building_type** column has 55 unique categories, which is very high for 150 records. Many categories represent the same concept with slightly different naming conventions. For example, 'Office', 'Office / Commercial', 'Office building', 'Office, Retail', and 'commercial office' all refer to office buildings. Similarly, 'Mixed use', 'Mixed-use', 'Mixed', and 'Mixed-use : Residential / Commercial' are variations of mixed-use buildings. This indicates a data format inconsistency that could be resolved by standardizing category names.

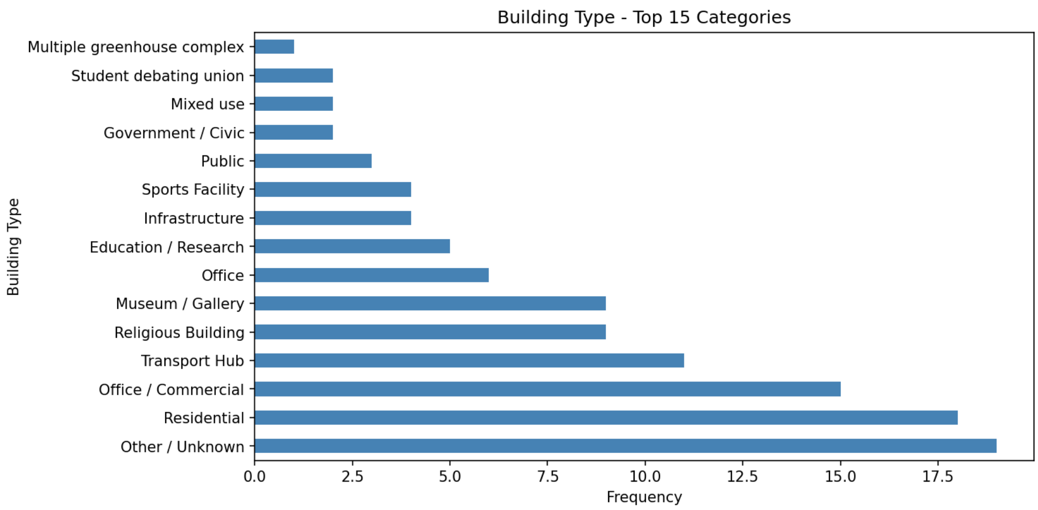

Figure 1: Top 15 building type categories. 'Other / Unknown' and 'Residential' are the most frequent.

3. Missing Data Analysis

3.1 Missing Data Rate

Three columns have missing values: **year** has the highest missing rate at 49.33% (74 out of 150 records), followed by **location** at 28.67% (43 records), and **coordinates** at 9.33% (14 records). The remaining columns (building_name, architect, building_type, wiki_url) have no missing values.

Column	Missing Count	Missing Rate (%)
building_name	0	0.00
architect	0	0.00
location	43	28.67
year	74	49.33
building_type	0	0.00
coordinates	14	9.33
wiki_url	0	0.00

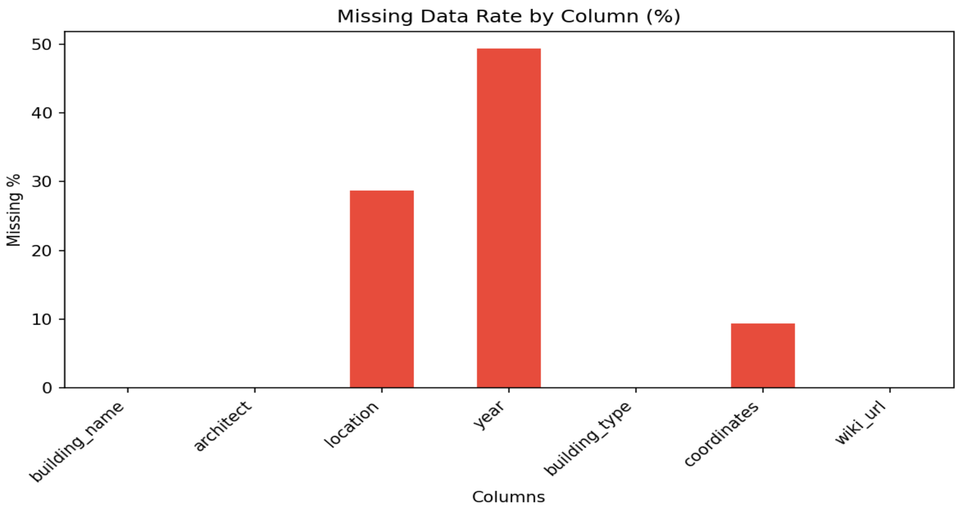

Figure 2: Missing data rate by column. Red bars indicate columns with missing values.

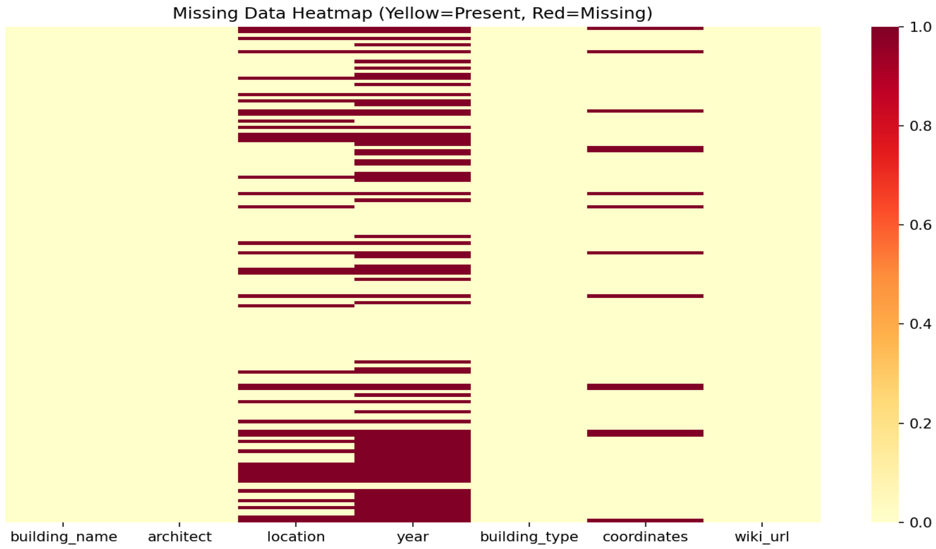

Figure 3: Missing data heatmap showing the pattern of missing values across all rows and columns.

3.2 Missing Data Type (MCAR, MAR, MNAR)

year (49.33% missing): Construction year information is often unavailable for older or lesser-known buildings on Wikipedia. Since the missingness is likely related to the building's age and notability (unobserved characteristics), this is classified as **MNAR** (Missing Not At Random).

location (28.67% missing): Location data depends on the structure and completeness of the Wikipedia page. Some Wikipedia infoboxes do not include a location field. Since the missingness depends on the page format (an observed variable), this is classified as **MAR** (Missing At Random).

coordinates (9.33% missing): Coordinate data is typically extracted from Wikipedia infoboxes. Its absence depends on whether the page has a geo-tagged infobox, making this **MAR** (Missing At Random).

4. Outlier Analysis (IQR Method)

The IQR (Interquartile Range) method was applied to the only numeric column, **year**. Q1 = 1981.5, Q3 = 2012.5, IQR = 31.0. The lower bound is 1935.0 and the upper bound is 2059.0. Four outliers were detected with values: **1918, 1928, 1930, and 1931**. These represent buildings constructed significantly earlier than the majority of the dataset, which is concentrated in the 1980-2020 range.

Metric	Value
Q1 (25th percentile)	1981.5
Q3 (75th percentile)	2012.5
IQR	31.0
Lower Bound (Q1 - 1.5*IQR)	1935.0
Upper Bound (Q3 + 1.5*IQR)	2059.0
Number of Outliers	4
Outlier Values	1918, 1928, 1930, 1931

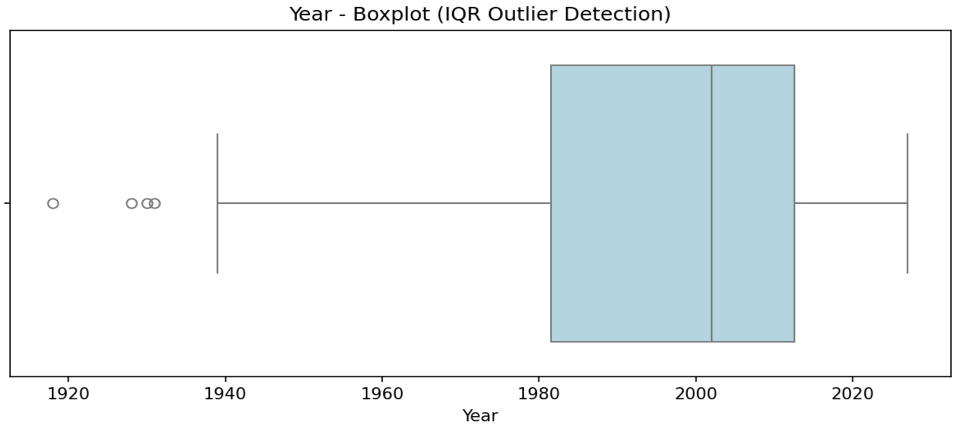

Figure 4: Boxplot of the year column showing the distribution and outliers on the left side.

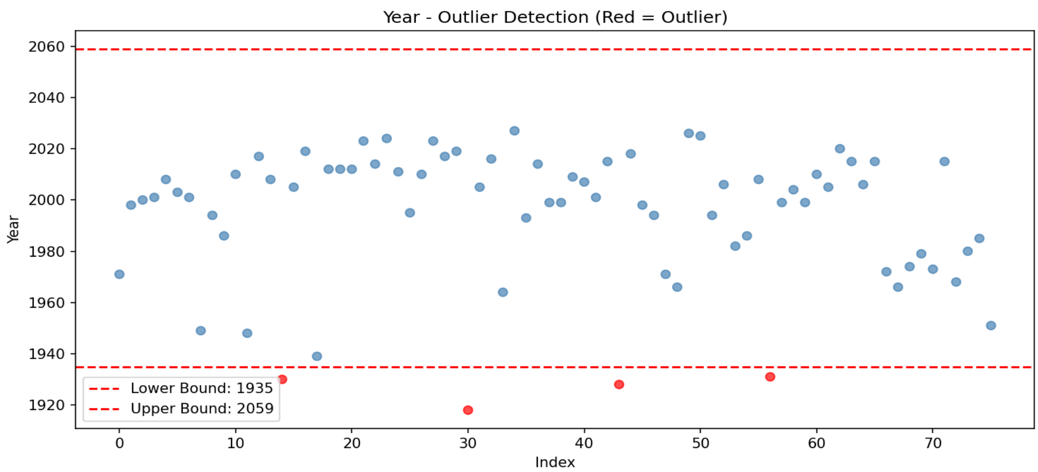

Figure 5: Scatter plot showing outliers in red. The dashed lines represent the IQR boundaries.

5. Distribution and Statistical Summary

Statistical analysis was performed on the **year** column (the only numeric variable). Note: Skewness and Kurtosis are not calculated for categorical variables as per the assignment requirements.

Statistic	Value	Interpretation
Mean	1994.67	Average construction year
Median	2002.00	Middle value (50th percentile)
Mode	1999.0	Most frequent year
Std Deviation	25.79	Spread of years
Skewness	-1.20	Left-skewed (tail towards older years)
Kurtosis	0.77	Slightly leptokurtic

The negative skewness (-1.20) indicates that the distribution is left-skewed, meaning there are more buildings with recent construction years and a longer tail towards earlier years. The mean (1994.67) being lower than the median (2002.0) confirms this left skew. The kurtosis value (0.77) indicates a slightly peaked distribution compared to a normal distribution.

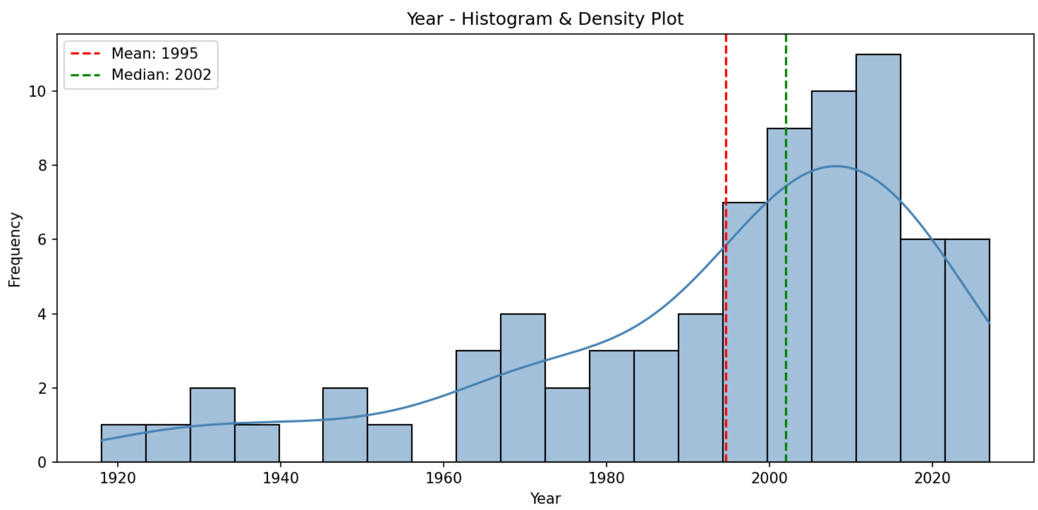

Figure 6: Histogram with density plot for the year column. Red dashed line = Mean, Green dashed line = Median.

6. Repeated Data Check

A duplicate check was performed across all columns. The result shows that there are **0 duplicate records** in the dataset. Each row represents a unique building entry. This is expected since each building has a unique Wikipedia URL and building name.

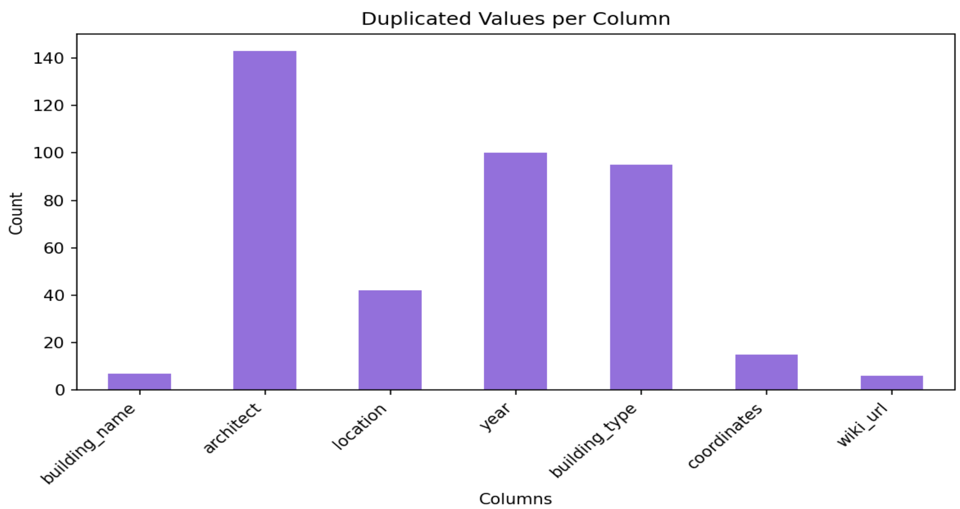

Figure 7: Duplicated values per column. The architect column has the most repeated values (143) since multiple buildings share the same architect. Building names and wiki URLs have very few repeats.

7. Bias Check

Bias analysis was conducted to evaluate whether categorical groups are balanced in the dataset.

7.1 Architect Distribution

The architect distribution shows a significant imbalance. **Zaha Hadid** accounts for 28.67% of all records (43 buildings), while **Berthold Lubetkin** has only 1 record (0.67%). Richard Rogers (18.67%), Alfred Waterhouse (18.0%), and Nicholas Grimshaw (16.67%) also have substantial representation. This imbalance could introduce bias if the dataset is used for modeling, as results would be disproportionately influenced by Zaha Hadid's buildings.

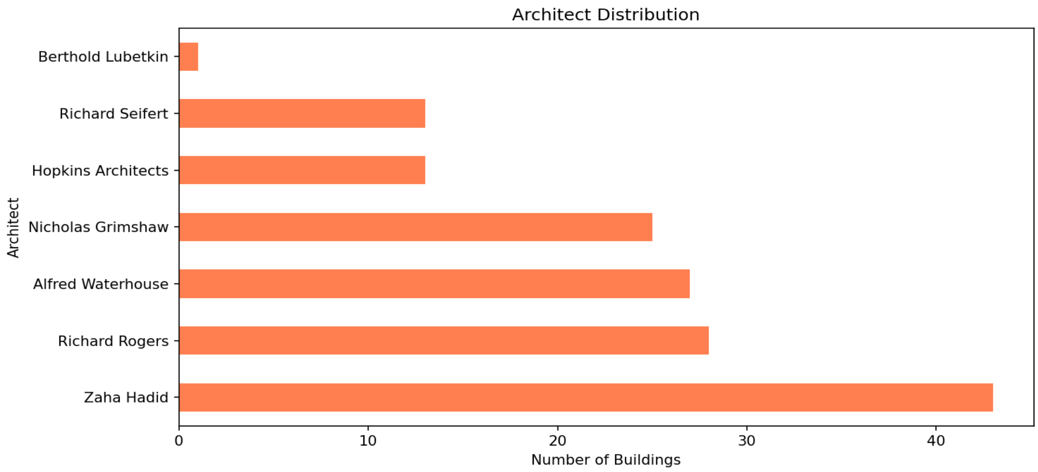

Figure 8: Architect distribution showing Zaha Hadid dominating the dataset.

7.2 Building Type Distribution

The building_type column has 55 unique categories for 150 records, resulting in many categories with only 1 entry (0.67%). The top categories are 'Other / Unknown' (12.67%), 'Residential' (12.0%), and 'Office / Commercial' (10.0%). This high cardinality and long-tail distribution indicate significant representation imbalance. Most building types appear only once, making statistical analysis per category unreliable.

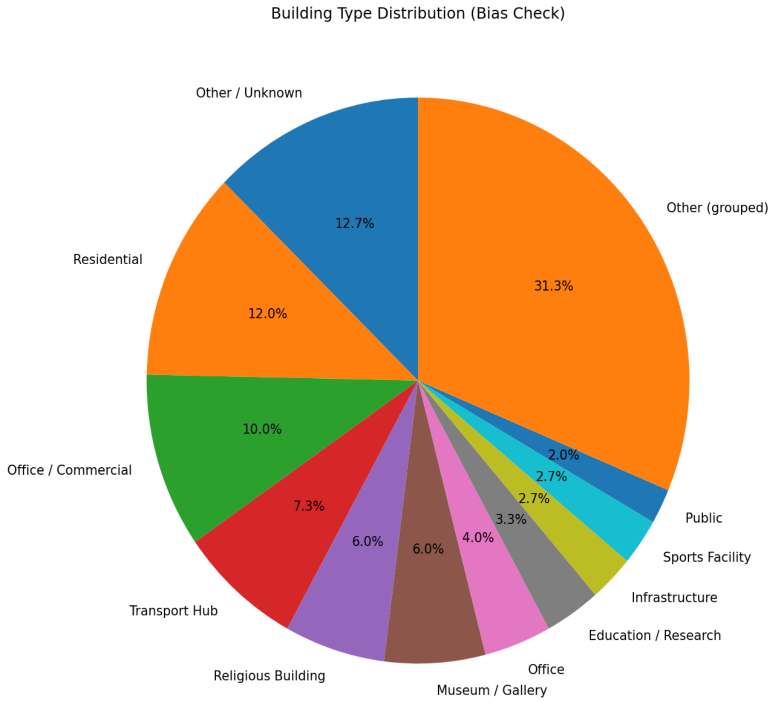

Figure 9: Building type distribution pie chart. Categories with fewer than 3 entries are grouped as 'Other'.

8. Bonus: Semantic Outlier Analysis

A semantic outlier analysis was performed using TF-IDF vectorization and cosine distance. All text columns (building_name, architect, location, building_type) were combined into a single text field, then vectorized. The average cosine distance from each record to all other records was calculated. Records above the 90th percentile threshold were flagged as semantic outliers.

A total of **15 semantic outliers** were detected. These are records whose combined text content is most dissimilar from the rest of the dataset. Examples include infrastructure projects (bridges, airports) and buildings in unusual locations that differ from the dominant patterns in the data.

Building Name	Type	Architect
Enneus Heerma Bridge	Infrastructure	N. Grimshaw
Neville Bonner Bridge	Infrastructure	N. Grimshaw
Pulkovo Airport	International	N. Grimshaw
Zurich Airport	Public	N. Grimshaw
Danjiang Bridge	Infrastructure	Zaha Hadid
Jockey Club Innovation Tower	Education / Research	Zaha Hadid
Sky Park (Bratislava)	Mixed-use	Zaha Hadid
Maharashtra Cricket Assoc. Stadium	Sports Facility	Zaha Hadid
Heathrow Terminal 5	Airport terminal	R. Rogers
Madrid-Barajas Airport	Airport terminal	R. Rogers

Table: Top 10 semantic outliers detected by TF-IDF + cosine distance analysis.

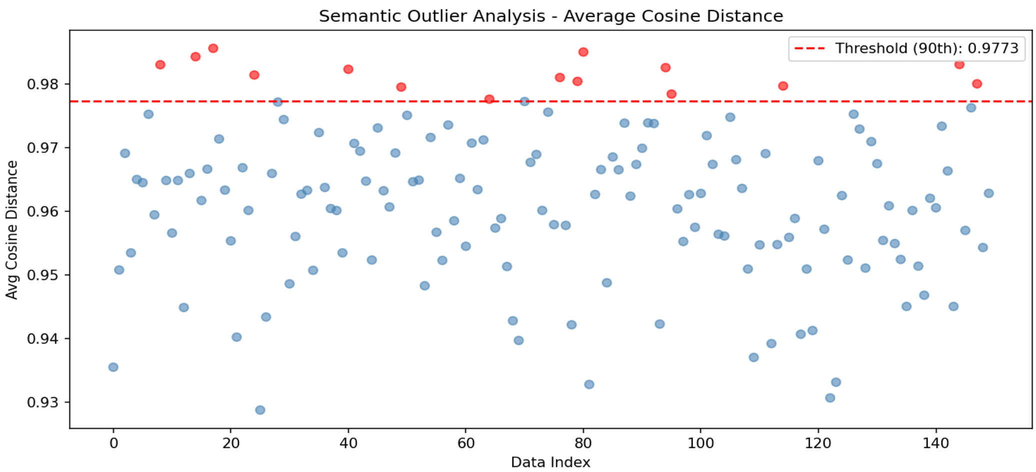

Figure 10: Semantic outlier scatter plot. Red points are above the 90th percentile threshold.